Chapter 4 Activity: Model Your System Like an Analyst

1. System Requirements Checklist

A. Outputs *(What the system produces for users)*

Sales Reports – Generate daily, weekly, and monthly summaries of transactions and profits.
Inventory Reports – Display available, sold, and repossessed vehicles for stock monitoring.
Customer Statements – Summarize each customer's purchase history, payments, and balances.
Repair Service Reports – Present job orders, repair status, costs, and mechanic performance.
Document Archive Logs – Record uploads, retrievals, and user access history for documents.
System Notifications – Alert users of expiring documents, maintenance schedules, and pending tasks.
Admin Dashboard – Show real-time dealership statistics, sales summaries, and user activity.

B. Inputs *(What the system receives or collects from users)*

Customer Information Forms – Collect customer details such as name, contact, and payment method.
Vehicle Registration Data – Record vehicle brand, model, plate number, chassis, price, and condition.
Repair Service Requests – Input details of vehicle issues, mechanic assignments, and estimated costs.
Sales Transaction Forms – Capture sales data including car sold, salesperson, and customer ID.
Document Upload Forms – Accept OR/CR, insurance, photos, and other car-related files.
Login Credentials – Require username and password for system access.
Feedback Forms – Gather customer ratings and service feedback for quality assessment.

C. Processes *(What the system performs internally)*

Inventory Management – Add, update, and remove vehicles from stock; track car availability.
Sales Processing – Create invoices, compute totals, and record all completed transactions.

Customer Management (CRM) – Maintain customer profiles, contact history, and sales records.
Repair and Service Tracking – Record job orders, assign mechanics, and compute service costs.
Document Archiving and Retrieval – Store and retrieve uploaded vehicle papers securely.
Report Generation – Compile data into summarized dashboards and downloadable reports.
Authentication and Authorization – Verify login credentials and assign user permissions.

D. Performance Requirements

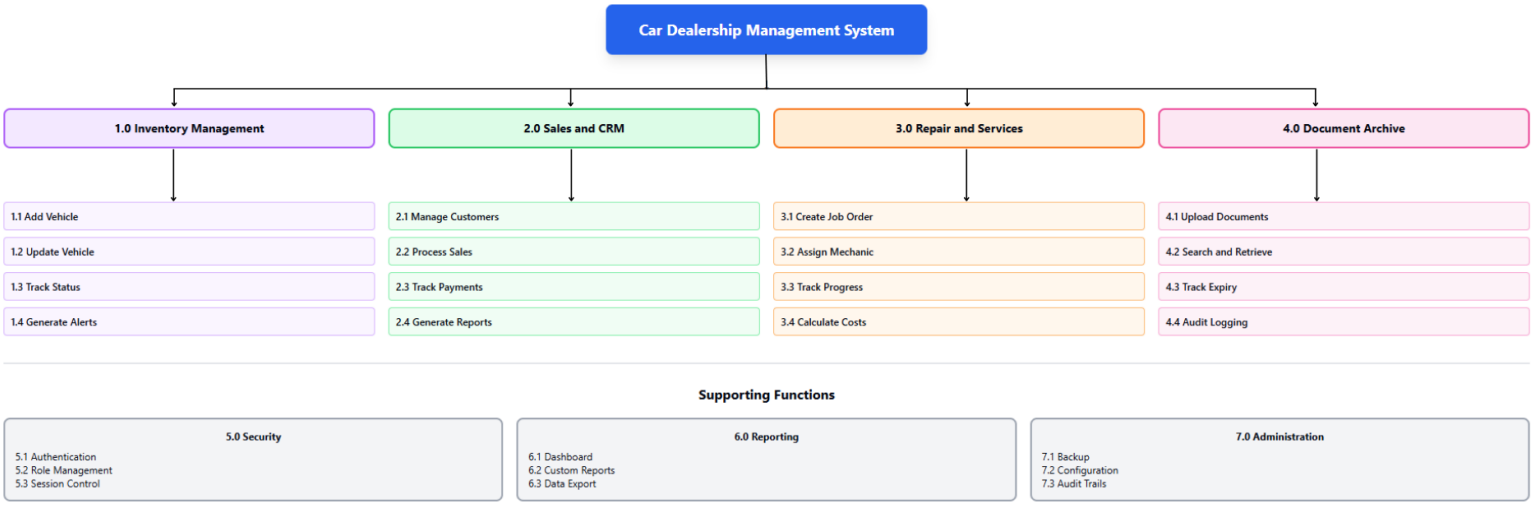
System Response Time – Load pages and process transactions within three seconds.
Concurrent Users – Support at least 50 simultaneous active users.
Data Storage Capacity – Handle over 10,000 records including cars, customers, and documents.
System Availability – Maintain 99% uptime during business hours.
Backup Frequency – Perform automatic daily backups with weekly archive copies.

E. Controls

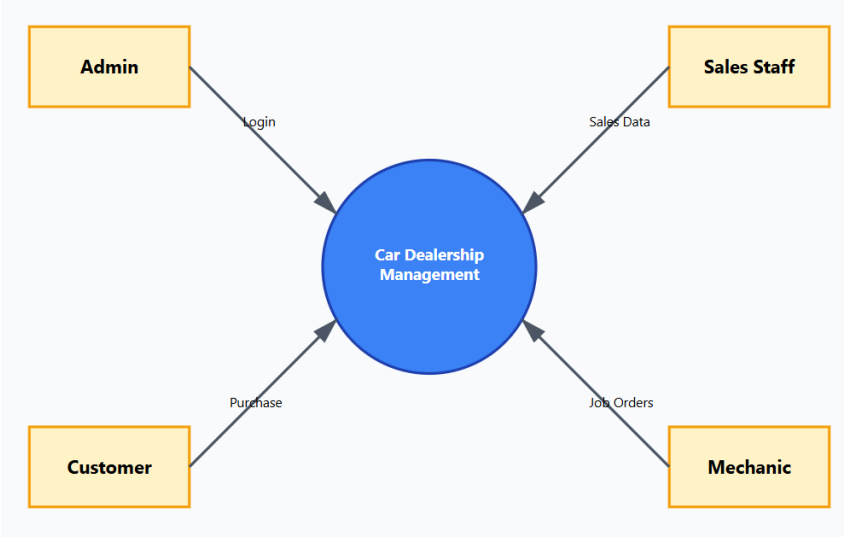
User Authentication – Require valid credentials for all users before accessing modules.
Role-Based Access Control – Grant access according to role (admin, sales, mechanic).
Data Encryption – Protect stored data and uploaded documents with encryption.
Input Validation – Check data accuracy and format before saving.
Audit Trail Logging – Record critical user actions such as edits, uploads, and deletions.
Session Timeout – Log out inactive users after a set period of time.
Backup and Recovery – Automatically back up data and test recovery procedures regularly.

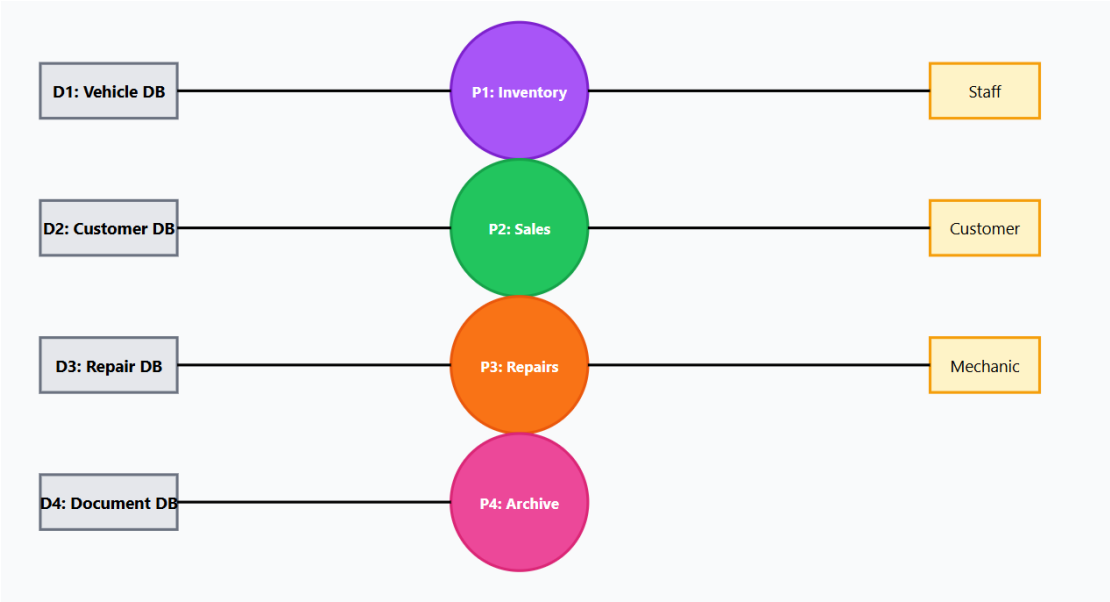
2. Create Modeling Diagrams

Functional Decomposition Diagram (FDD)



Data Flow Diagram (DFD)





3. Total Cost of Ownership (TCO)

The Elmario Dealer Management & CRM project involves several direct and indirect costs over its six-month development and deployment phase. The figures below reflect realistic costs for a medium-complexity web system using PHP, Symfony, Twig, and Tailwind CSS.

Direct Costs

- **Developer Salary** – Payment for system design, coding, testing, and deployment over six months.
- **Software Licenses and Tools** – Subscription or one-time fees for IDEs, frameworks, and plugins used in development.
- **Server Hosting & Domain** – Web hosting services and domain name registration for one year.

TCO Summary Table

Cost Category	Description	Estimated Cost
Developer Salary	System analysis, coding, testing, and deployment	₱ 180,000
Software Licenses & Tools	IDEs, paid libraries, or productivity software	₱ 20,000
Server Hosting & Domain	Web hosting and domain for one year	₱ 12,000
Training & Orientation	Training sessions for staff and mechanics	₱ 8,000
Maintenance & Support	Regular post-launch updates and bug fixes	₱ 15,000/year
Operational Downtime	Productivity loss during training and migration	₱ 10,000